

HW4: FIR filter design via the window method

Due 23:59 on Sunday May 3, 2026

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Part I: Hand-written exercises

1. **(40 pts). Low-pass to high-pass transformation.** Suppose that $H(z) = B(z)/A(z)$ denotes the system function of an IIR low-pass filter. As usual, we assume that the system is causal and stable, and the corresponding difference equation is denoted as $y[n] = -\sum_{k=1}^N a_k y[n-k] + \sum_{k=0}^M b_k x[n-k]$.
 - (a) Show that $H(-z)$ has a high-pass magnitude response.
 - (b) How do we modify the coefficients a_k 's and b_k 's so that the resulting difference equation has a high-pass response?
 - (c) Continuing from above, if the low-pass filter's cutoff frequency is $\pi/5$, what is the cutoff frequency of the high-pass filter after your transformation?
 - (d) Discuss how to design a band-pass filter using combination of low-pass and high-pass filters.

2. **(30 pts) Bilinear transformation for all-pass filtering.** We have shown in FUD3 that an all-pass filter $H(z) = \frac{1-qz^{-1}}{1-pz^{-1}}$ where $p = re^{j\theta}$ and $q = \frac{1}{p^*} = \frac{1}{r}e^{j\theta}$ has one zero and one pole at reciprocal locations. When $r < 1$, $H(z)$ represents a causal and stable filter. What we have not shown is that the filter has a positive group delay at every frequency. It turns out that this is easier to show via inverse bilinear transformation.
 - (a) Let $z(s) = (1+s)/(1-s)$. Define $H_c(s) = H(z(s))$. You will find that $z(s)$ has a pole at $s_p = u_p + iv_p$ and a zero at $s_q = u_q + iv_q$. Show that $u_p = -u_q < 0$ and $v_p = v_q$.
 - (b) Using geometric interpretation, show that $\partial \angle H_c(j\Omega)/\partial \Omega < 0$ for all Ω .
 - (c) Show that $\partial \angle H(e^{j\omega})/\partial \omega < 0$ for all ω . [Hint: $\Omega(\omega)$ increases monotonically].

Part II: Computer-based exercises

Please turn in this part as if you are writing a lab report. Instead of answering each item (a),(b),(c)... separately, you may organize your report freely. The page limit is 2.

- 3. (30 pts) FIR filter design to meet specs.** Suppose that we are again asked to design a low-pass filter with edge frequency $\omega_p = \pi/3$, DC gain $|H(e^{j0})| = 1$, and a requirement that the gain drops below 0.01 at $\omega_s = 0.4\pi$, i.e. $|H(e^{j(0.4\pi)})| \leq \delta = 0.01$, which corresponds to -40 dB. Also, we require the passband ripple to be no more than 0.01. These specifications are similar as the previous homework, but we are going to design an FIR filter this time.
- (a) From Table 7.2 of the textbook, we know that the Hann window has a peak approximation error below -40 dB. Find the minimum length of the filter to meet the transition width requirement.
 - (b) Now, fix $\delta = 0.01$, but instead, vary ω_s between 0.4π and 0.5π . Investigate how the required order N varies.
 - (c) Let's compare the FIR filter that you just designed and the Butterworth filter you obtained last time. When using them to filter a signal, which one is computationally more efficient?